

Rethinking Optical Flow Methods for Micro-Expression Spotting

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ABSTRACT

Micro-expressions (MEs) spotting is popular in some fields, for example, criminal investigation and business communication. But it is still a challenging task to spot the onset and offset of MEs accurately in long videos. This paper refines every step of the workflow before feature extraction, which can reduce error propagation. The workflow takes the advantage of high-quality alignment method, more accurate landmark detector, and also more robust optical flow estimation. Besides, Bayesian optimization hybrid with Nash equilibrium is constructed to search for the optimal parameters. It uses two players to optimize two types of parameters, one player is used to control the ME peak spotting, and another for optical flow field extraction. The algorithm can reduce the search space for each player with better generalization. Finally, our spotting method is evaluated on MEGC2022 spotting task, and achieves F_1 -score 0.3564 on CAS(ME)³-UNSEEN and F_1 -score 0.3265 on SAMM-UNSEEN.

CCS CONCEPTS

• Human-centered computing

[†] These authors are equal contribution to this work.

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KEYWORDS

Micro-expressions, Optical flow method, Bayesian optimization, Characteristic curve

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1 INTRODUCTION

Expressions reflect the changes of human emotions, which can be divided into macro-expressions (MaEs) and MEs according to their intensity and duration. MEs have short duration and low intensity. Because of these properties, the progress in research related to MEs is very slow.

The facial movements are divided into habitual expressions, MaEs, or MEs. Overall, MaEs and MEs differ significantly in duration and intensity: in duration, MaEs usually last from 0.5s to 4s, while MEs often less than 0.5s; in intensity, the facial movements of MEs are small and imperceptible. These differences make the precise spotting of MEs more difficult.

This paper refines the steps of workflow, for example, face alignment, regions of interest (ROIs) generation and feature extraction. To achieve high-quality alignment, intermediate indices use floating point format to reduce the cumulative error, which can make ROIs consistent in different people. Therefore, the movement calculated from optical flow field is more accurate. Furthermore, this paper also uses a more accurate landmark detector and robust optical flow estimation, which can help us to control the error from the source.

As everyone knows that excellent parameters make the model work outstanding. But searching the parameters is expensive. We propose a Bayesian optimization hybrid with Nash equilibrium algorithm, which uses two players to optimize two types of parameters at the same time, and updates them with momentum. The Nash game strategy reduces the searching time for parameters and improves the performance of the model.

2 RELATED WORKS

Spotting task can be categorized into two main categories: deep learning methods and non-deep learning methods. The first category uses deep learning methods to extract temporal and spatial information, or uses deeper networks to process features. The second category uses prior knowledge to extract features, which is more complex in terms of dealing with the temporal patterns of the video.

Some scholars have applied deep learning methods to study MEs spotting. Wang et al. [1] use 2+1D Spatiotemporal Convolutional Network to extract spatial and temporal features, and then use Clip Proposal Network to get some MEs clips. Yu et al. [2] introduce a pyramidal convolutional neural network in order to extract more complete feature information.

However, the robustness of deep learning methods usually can't be guaranteed when dealing with small datasets, and on the contrary, many non-deep learning methods perform more stable. For example, Li et al. [3] extract features from a projection in PCA space and use SVM to distinguish MEs from other facial movements. In addition, there are also some works of extracting the features of adjacent frames from optical flow [2,4]. These methods can improve the stability of the model based on the dataset.

3 METHODOLOGY

The spotting task of long-face video has two challenges: First, habitual facial movements (such as blinking) may cause false spotting of MaEs and MEs. Second, head movements incorporate overall translation and rotation, which will affect the performance of the optical flow method.

To address the challenges mentioned above, a modified algorithm is proposed in this paper. The proposed algorithm contains four main parts: (1) Preprocess the video. We use the sliding window of 200 frames to extract the features in the temporal dimension. (2) Face cropping and alignment. The algorithm proposed in this paper first acquires 68 landmarks, then we propose a method to generate face box, and use optical flow method to extract facial features and align the faces of images from adjacent frames. (3) Generate the ROIs and extract the feature. The generation of ROI is determined by action units (AUs). We count the AU information contained in the MaEs and MEs datasets to generate the corresponding ROIs, and then use optical flow methods to extract features within the ROIs. (4) Use Bayesian optimization method to search the parameter space to improve model performance.

3.1 Face Cropping and Alignment

Each frame of video needs to be extracted to crop out the region of face. After the region of face is resized to specified size, for example, 256-pixel, a further alignment algorithm will make the face between different frames strongly aligned.

Landmark is used to determine the face box, since the prediction of the face box is oscillating. The face box is generated according to the distance between the eyes.

$$p_{mid} = \frac{p_{le} + p_{re}}{2} \quad (1)$$

$$d_{ref} = x_{re} - x_{le} \quad (2)$$

Where $p_{le} = (x_{le}, y_{le})$ represents the landmark of the left eye, so does p_{re} . d_{ref} is the distance between the eyes.

$$P_{box} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} p_{mid} + d_{ref} \begin{bmatrix} \gamma_l & \gamma_d \\ \gamma_r & \gamma_u \end{bmatrix} \quad (3)$$

The P_{box} matrix stores the position information of the face box. γ_i is considered as a parameter of the corresponding position of the face box, which represents a multiple of the distance between the eyes, and (l, d, r, u) are four orientations of the face box.

The alignment algorithm has two steps. In the first step, the optical flow method is used to calculate the deviation between two frames. In view of the faces given in the dataset are nearly Full Frontal [5], we assume that the optical flow vector of the nose tip can represent the translation of the face. According to the assumption, the process of alignment uses only translation, and the translation of pixels introduces less error than rotation due to the error of linear interpolation.

$$\vec{U}_{face} := \vec{U}_{nose} = \frac{1}{S} \sum_{p=(x,y) \in M_{nose}} \vec{u}_p \quad (4)$$

Where $\vec{u}_p$ is the optical flow vector of point p . M_{nose} is region of the nose tip and S is the total points of the region. This paper uses $\vec{U}_{nose}$ instead of $\vec{U}_{face}$ to represent the optical flow vector of the face movement.

In the second step, the bias due to instability of the optical flow method is calculated to measure the effect of alignment. We keep the bias to 0.5-pixel or less. If it is greater than 0.5-pixel, step one is repeated. After strong alignment, the face box between subsequent frames and referenced frame is overlapped.

3.2 ROI Generation and Feature Extraction

Facial Action Coding System (FACS) is a system originally developed by Paul Ekman and Wallace Friesen in 1976, to taxonomies every conceivable human facial expression [6]. Based on the anatomical characteristics of the face, they defined Action Unit (AU) and analyzed the motor characteristics of these AUs. (Such as "Inner Brow Raiser", "Brow Lower", "Upper Lip Raiser" etc.). The histogram of AU occurrence of CAS(ME)² is shown in figure1 based on the first 45 AUs in abscissa.

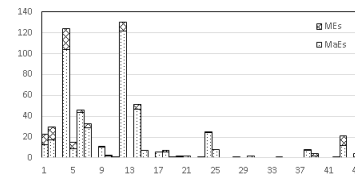


Figure 1: The first 45 AUs count in CAS(ME)²

AU1, AU2 and AU4 represent the movements around the eyebrows. Figure 1 shows the occurrence of expressions, the horizontal coordinates indicate the different AU areas. According to figure 2, this paper constructs the ROIs around the eyebrows, nose and mouth. The generation of ROIs in our algorithm depends on 68 landmarks.

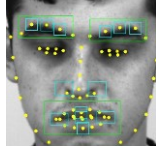


Figure 2: Facial landmarks and ROIs

After the process of ROIs generation is completed, the optical flow method is then used in this paper to obtain the optical flow field between the sequential frames and the reference frame. Other ROIs feature is exacted as the same as the region of the nose.

$$\vec{U}_i = \frac{1}{S} \sum_{p=(x,y) \in M_i} \vec{u}_p \quad (5)$$

Where, M_i is a specified region. In MEs spotting, the intensity of the acquired optical flow field information is very low. In this paper, the first n percentile of optical flow values is chosen to represent the motion of a region, which can amplify the motion information.

$$\vec{U}_{i,n\%} = \frac{1}{S} \sum_{p=(x,y) \in M_{i,n\%}} \vec{u}_p \quad (6)$$

Finally, for there still exist some pixels of bias, which are less than 0.5, the features of ROI still need to be aligned with $\vec{U}_{face}$ to eliminate the effect of the bias. The process is described by (7):

$$r_i = |\vec{U}_{i,n\%}| - |\vec{U}_{face}| \quad (7)$$

Where r_i is the relative movement between current frame and reference frame.

Except for the reference frame, a (16×2) matrix Ψ is generated for each image frame. The matrix Ψ can be expressed as (8):

$$\Psi = [r_{i,j}] \quad (8)$$

Where i indicates specify of region, and j indicates the dimension of the optical flow field.

3.3 Expression Spotting Method

Based on the conclusions above, we use signal processing to enhance the expression of facial features. When an expression occurs, the facial feature curve produces a distinct peak. By looking for the peak of the feature curve in each window, the time period in which the expression occurs can be spotted.

Since the task needs to obtain the onset frame [8] and offset frame of the expression, after locating the peak, we extend the wave peak to both sides.

The spotting of wave peak is based on the sliding windows. This paper proposes a sliding windows strategy which performs well on spotting task.

The span of slide window is fixed at 200 frames. The first frame is taken as the reference frame, and the other frames are considered as the subsequent frames. In our strategy, the default step size is 100. It should be noted that, during the slide process, the reference frame sometime will correspond to the wave peak,

which will affect the time series data. To solve this problem, this paper moves the reference frame forward or backward to the trough of the feature curve as it moves near the crest of the wave.

3.4 Parameter Tuning

Bayesian optimization algorithm [7] is the mainstream method in parameter optimization. It can search the parameter approximations that maximum the objective function in a high-dimensional space with several steps. Optimization algorithms usually consist of many strategies. Nash equilibrium strategy is a method which can find more robust parameters, for it can help a parameter to determine the best payoff in a situation not only based on their decisions but also based on the decisions of other parties involved [8,9]. The whole procedure of Bayesian algorithm hybrid with Nash equilibrium can be outlined as follows.

Bayesian Optimization Hybrid with Nash Equilibrium

Input: $\mathcal{D}$ is a video dataset

1: $\theta^{(1)} = [\theta_1^{(1)}, \theta_2^{(1)}, \dots, \theta_N^{(1)}]$ is the parameter for searching, and is split into two groups $\theta^{(1)} = [\theta_{g1}^{(1)}, \theta_{g2}^{(1)}]$

2: Maximum the objective function $G_{d \in \mathcal{D}}(F(d, \theta_{g1}, \theta_{g2}), L_{GT})$, F is the prediction function for MEs, L_{GT} is the ground truth of MEs

3: for $t \in 1, \dots, T$ do

4: Fix $\theta_{g1} = \theta_{g1}^{(t)}$ and search the best performance of $\theta_{g2}^{(t)}$ by Bayesian Optimization

5: Let $\theta_{g2}^{(t+1)} = \theta_{g2}^{(t)} + \alpha \theta_{g2}'^{(t)}$

6: Fix $\theta_{g2} = \theta_{g2}^{(t)}$ and search the best performance of $\theta_{g1}^{(t)}$ by Bayesian Optimization

7: Let $\theta_{g1}^{(t+1)} = \theta_{g1}^{(t)} + \alpha \theta_{g1}'^{(t)}$

8: end for

9: return $\theta^{(T)}$

Parameters in the paper are split into two groups for tuning due to the complexity of the parameter space. One group of parameters is used to spot ME peak, and the other group is used for optical flow field extraction. Then, two sets of parameters can be set to optimal values by Bayesian optimization algorithm hybrid with Nash equilibrium strategy.

4 EXPERIMENTS AND RESULTS

4.1 Evaluation and Datasets

All experiments are performed on SAMM-LV [10] and CAS(ME)² [11], tested on SAMM-UNSEEN [12] and CAS(ME)³-UNSEEN [13] datasets. The fps of SAMM-LV and SAMM-UNSEEN is 200, while the fps of CAS(ME)² and CAS(ME)³-UNSEEN is 60. In the experiment, the IOU of the detection interval and the ground-truth interval is calculated by the (9).

$$\text{IOU} = \frac{W_{\text{predict}} \cap W_{\text{groundTruth}}}{W_{\text{predict}} \cup W_{\text{groundTruth}}} \quad (9)$$

It should be noted that, if the IOU is greater than 0.5, the prediction is considered as a true positive result (TP), otherwise, it will be treated as a false positive result (FP). Expression intervals which are not predicted in ground-truth will be treated as a false negative result (FN). The performance of the spotting method is evaluated by the F1-score, which is defined as:

$$\text{F}_1\text{-score} = \frac{2 * TP}{2 * TP + FP + FN} \quad (10)$$

4.2 Hyperparameter Settings

Both SAMM-LV and CAS(ME)² are uniformly sampled to 30fps to improve the efficiency of MEs spotting, for there are many redundant frames in the MEs datasets. In order to further remove some distracting areas of the face, the face is cropped into a smaller image that encloses all ROIs.

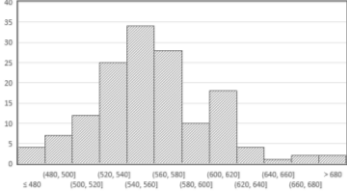


Figure 3: Histogram of cropped face size on CAS(ME)²

In this experiment, $\gamma_d, \gamma_w, \gamma_l, \gamma_r$ are set to -1.5, 2.5, -2, 2, respectively, and the cropped face image is resized to 256×256 pixel. Figure 3 shows the face cropping size which is set to $4d_{ref}$.

4.3 Landmark Detection Experiments

Dlib[14] and MobileFaceNet-based models are evaluated on CAS(ME)². As shown in Figure 4, the left side is the facial landmarks detected by the Dlib, and the right side is the facial landmarks detected by the MobileFaceNet-based [15] models. It is also noted that Dlib will fail in some scenes with glasses, and MobileFaceNet-based models are more robust.

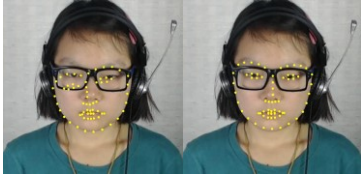


Figure 4: Facial landmarks by different detector (©Xiaolan Fu)

Table 1 shows the ME spotting results on CAS(ME)² using our MobileFaceNet-based models and using the Dlib. The F1-score of ME spotting will improve as the accuracy of landmarks detection.

Table 1: The F1-score of different landmark detection methods on CAS(ME)²

	MaEs	MEs	Overall
Dlib	0.4403	0.2269	0.4070
Ours	0.4547	0.2461	0.4232

4.4 Optical Flow Estimation Experiments

The optical flow method plays a crucial role in MEs spotting, hence nine optical flow estimation methods are constructed to tested on CAS(ME)² to verify their performance. All optical flow estimation methods are tested in the same environment with the same parameters.

The experiment results are shown in Table 2, it can be seen that the ME spotting is very sensitive to optical flow feature extraction methods, and among all the methods tested, the TV-L1 optical flow estimation method works the best, and hence, TV-L1 optical flow estimation is finally taken as the method used in this paper.

Table 2: Comparison of different optical flow methods on CAS(ME)²

method	Overall			MEs		
	precision	recall	F1	precision	recall	F1
Farneback [16]	0.243	0.507	0.328	0.096	0.246	0.138
Simpleflow [17]	0.226	0.493	0.310	0.091	0.263	0.135
Deeflow [18]	0.089	0.377	0.144	0.032	0.105	0.050
Disflow [19]	0.281	0.535	0.368	0.1	0.228	0.139
flowNet2.0 [20]	0.022	0.249	0.041	0.024	0.290	0.044
LiteFlowNet2.0 [21]	0.02	0.341	0.037	0.024	0.397	0.046
PWC-Net [22]	0.024	0.445	0.046	0.031	0.493	0.059
MaskFlowNet [23]	0.026	0.445	0.050	0.036	0.497	0.066
TV-L1 [24]	0.285	0.521	0.369	0.132	0.246	0.172

4.5 Final Results

Table 3 compares the F1-score of our model against He et al. [4] on CAS(ME)² and SAMM-LV. It can be seen that our model has better performance in expressions spotting.

Table 3: Comparison of F1-score of different spotting methods on CAS(ME)² and SAMM-LV

	CAS(ME) ²	SAMM-LV
He et al. [4]	0.3436	0.3638
Ours	0.4030	0.3863

Table 4 shows the performance of our model and baseline [25] on MEGC2022-testSet, which contains two parts, CAS(ME)³-UNSEEN and SAMM-UNSEEN, respectively. Our model achieves better performance than baseline on both datasets.

Table 4: Comparison of F1-score of different spotting methods on MEGC2022-testSet

	CAS(ME) ³ -UNSEEN	SAMM-UNSEEN
Baseline [25]	0.1739	0.1176
Ours	0.3564	0.3265

5 CONCLUSIONS

This paper rethinks the importance of optical flow, which is used both in alignment and feature extraction. In the step of alignment, robust optical flow estimation can reduce the error of translation, which is used in the correction of ROI movement. In the step of feature extraction, it guarantees the accuracy of the position series curve effectively. Inspired by the error control of optical flow estimation, each step of the workflow is refined to reduce the cumulative error. In terms of source, a better landmark method is introduced into the workflow. In terms of process, the intermedia indices are switched to floating point values from integer values. This paper also proposes a Bayesian optimization algorithm hybrid with Nash equilibrium, which uses two players to optimize the two types of parameters synchronously, and updates them with momentum. It greatly helps to save computational costs in the search parameters.

The proposed spotting method is performed on CAS(ME)² and SAMM-LV achieving F1-score at 0.4232 and 0.3863, respectively. It is worth mentioning the method is also generalizable on unseen datasets. The method achieves F1-score at 0.3564 on CAS(ME)³-UNSEEN and achieves F1-score at 0.3265 on SAMM-UNSEEN.

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